Hope Artificial Intelligence

Classification Assignment Problem Statement or Requirement:

A requirement from the Hospital, Management asked us to create a predictive model which will predict the Chronic Kidney Disease (CKD) based on the several parameters. The Client has provided the dataset of the same.

1.) Identify your problem statement:

-A per client request, need to create a best model which will predict the CKD based on the several parameters.

2.) Tell basic info about the dataset (Total number of rows, columns)

-399 rows × 25 columns

3.) Mention the pre-processing method if you're doing any (like converting string to number – nominal data)

-The pre-processing method is MinMaxScaler

4.) Develop a good model with good evaluation metric. You can use any machine learning algorithm; you can create many models. Finally, you have to come up with final model.

[13]:
```
from sklearn.metrics import f1_score
f1_macro=f1_score(y_test,grid_predictions,average='weighted')
print("The f1_macro value for best parameter {}:".format(grid.best_params_),f1_macro)
```

```
The f1_macro value for best parameter {'classifier': GaussianNB(), 'classifier__var_smoothing': 1e-08}: 0.9775556904684072
```

5.) All the research values of each algorithm should be documented. (You can make tabulation or screenshot of the results.)

```python
from sklearn.model_selection import GridSearchCV
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import MinMaxScaler
from sklearn.naive_bayes import GaussianNB, MultinomialNB, BernoulliNB

# 1. Create a pipeline with a MinMaxScaler to support MultinomialNB
pipeline = Pipeline([
    ('scaler', MinMaxScaler()),
    ('classifier', GaussianNB()) # Placeholder
])

# 2. Define the search grids for all three Naive Bayes models
param_grid = [
    {
        'classifier': [GaussianNB()],
        'classifier__var_smoothing': [1e-9, 1e-8, 1e-7]
    },
    {
        'classifier': [MultinomialNB()],
        'classifier__alpha': [0.1, 0.5, 1.0, 2.0]
    },
    {
        'classifier': [BernoulliNB()],
        'classifier__alpha': [0.1, 0.5, 1.0, 2.0]
    }
]
```

6.) Mention your final model, justify why u have chosen the same.

-Based on three parameters in grid search, gaussian is the best model with 0.98 accuracy.

Note: Mentioned points are necessary, kindly mail your document as well as .ipynb (code file) with respective name. ϖ ϖ Sub file name also should be properly named for Example (SVM_Ramisha_Assi-5.ipynb) Communication is important (How you are representing the document.) Kindly uploaded in the Github and Share it with us www.hopelearning.net Hope AI admin@hopelearning.net